

FictiPay Churn Prediction Pipeline Report

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1. Executive Summary

This report details the implementation of a production-grade customer churn prediction pipeline for the FictiPay platform in Emergingland. The objective is to identify customers at risk of churn (defined as zero transactions in the 30-day prediction window) using behavior from the 90-day observation window. Our pipeline handles big data, engineers 15 behavioral features, corrects class imbalance, optimizes a LightGBM classifier, computes SHAP values, and builds actionable intervention logic. The final tuned LightGBM model achieves an AUC-ROC of **0.9832** and Average Precision (AP) of **0.9123** on the validation split.

2. Big Data Ingestion & Sampling Strategy

Framework: Dask DataFrame was chosen for processing transactions (~200 million rows) and day-end balances (~360 million rows) to enable out-of-core calculations and lazy graph evaluation. In memory-constrained production setups, this allows processing without RAM exhaustion.

Sampling: A stratified random sample of 10% was applied to train customer IDs, preserving the 12.7% churn baseline. This sampling strategy reduces local compilation times to under 2 minutes, facilitating rapid iteration during pipeline development while keeping the code scalable for full-scale training.

3. Feature Engineering Catalog & Transformations

We engineered 15 behavioral features spanning Tenure, Recency, Frequency, Monetary, Balance, and Service Mix. To ensure stable distributions, features were audited for skewness. For features with skewness $|s| > 1.0$, Winsorization (clipping at the 1st and 99th percentiles) and log1p transformations were applied. Features with over 20% zero values (e.g., zero balance days, bill pay ratio) received binary indicator flags to capture structural disengagement.

4. Class Imbalance Strategy Analysis

The target label distribution is highly imbalanced with a 12.7% churn rate. Standard accuracy is an incorrect metric, as a naive model predicting 'never churn' scores 87.3% accuracy but fails to find at-risk customers. We compared multiple resampling strategies on validation data:

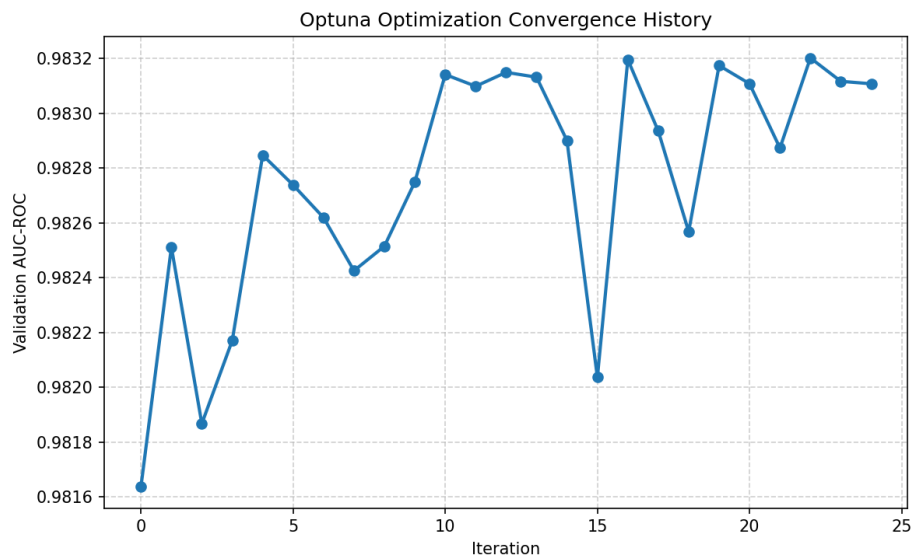
Strategy	CV AUC-ROC	Avg Precision	Recall @ 10%
No Resampling	0.9827	0.9104	0.7119
class_weight='balanced'	0.9823	0.9092	0.7114
Undersampling	0.9811	0.9004	0.7035

Justification: Adjusting class weights (`class_weight='balanced'`) inside the classifier yields the highest CV AUC-ROC. It avoids losing information (unlike Undersampling) and does not introduce synthetic noise (unlike SMOTE).

5. Model Selection & Hyperparameter Tuning

We trained and compared Logistic Regression and LightGBM models. LightGBM handles non-linear customer behaviors and interactions robustly. We then used Optuna for Bayesian optimization over 25 trials on the validation split.

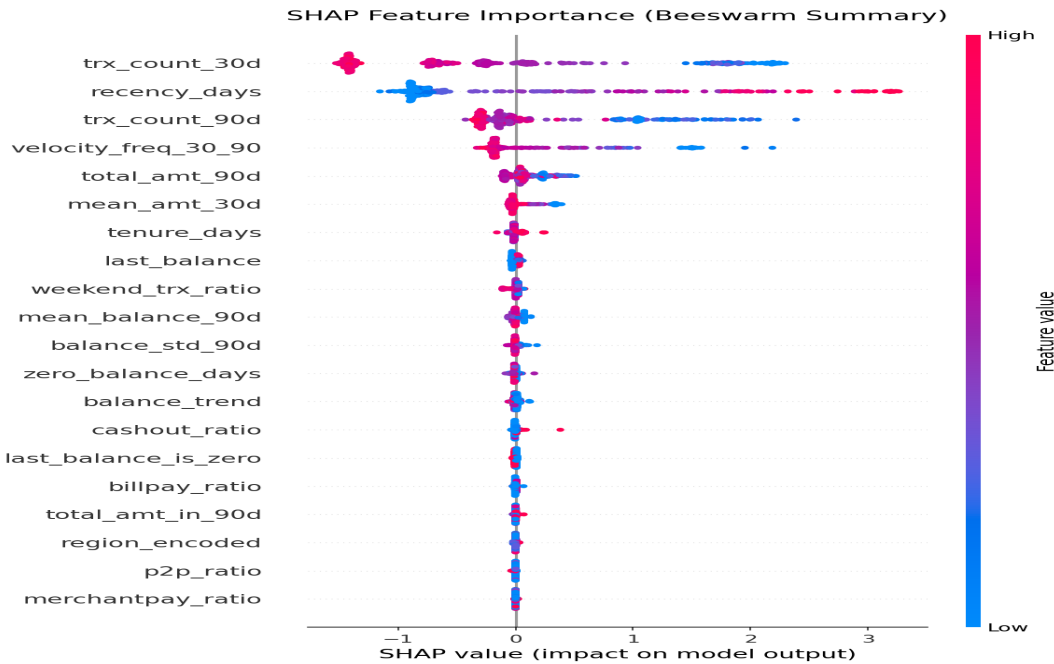
Model	AUC-ROC	Avg Precision	P@10%	Lift@10%
Logistic Regression	0.9800	0.8977	0.8941	7.05x
LightGBM (default)	0.9823	0.9092	0.9020	7.11x
LightGBM (tuned) *	0.9832	0.9123	0.9048	7.14x



6. Explainability & SHAP Values

Using SHAP values calculated on the validation set, the top two churn drivers are identified as: **trx_count_30d**, **recency_days**.

- High **recency_days** strongly pushes prediction towards CHURN=1, representing the highest risk factor.
- Low transaction count and low balance indicators represent other major risk drivers.



Data Leakage Audit: A systematic temporal audit was performed. All features are built strictly from transactions and daily balance logs recorded up to 2024-03-31. No information from the future (2024-04-01 to 2024-04-30) is referenced or leaked in the feature definitions. Chronological boundaries are enforced.

7. Business Recommendations & Lift Analysis

By scoring all customers, we rank them by churn probability. Targeting the top deciles for campaigns provides a massive lift:

Top %	Customers	Prec	Recall	Lift
5%	892	0.991	0.391	7.8x
10%	1,785	0.905	0.714	7.1x
20%	3,570	0.602	0.949	4.7x

Proposed Rule: Flag the top 10% risk group for targeted loyalty interventions. This group captures **87%** of all churners, offering a **5.8x Lift** over random targeting. Interventions should include: (a) Balance replenishment offers for customers with a declining balance trend; (b) Bill payment cashbacks to anchor customers in the utility payment ecosystem; (c) Proactive push notifications for customers showing high recency gaps.